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SAS Programming with an AI Assistant

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#SASinnovate

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Abstract

Explore how to effectively use ChatGPT as a programming partner when writing SAS® code. Learn best practices for prompting and validating results to accelerate your programming workflow. See how ChatGPT can assist with drafting, explaining, and refining SAS® programs.

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Exploring SAS Studio

1. Launch Google Chrome from the desktop and select **SAS Studio** from the Bookmarks bar.
2. Enter **student** in the **User ID** field and **Metadata0** in the **Password** field. Click **Sign in**. Select **Yes** when prompted to opt in to all assumable groups.
3. Click the **applications menu** icon in the upper left corner. The applications menu shows all the available applications. Depending on your environment, these applications might differ. Select **Develop Code and Flows** to open SAS Studio. We use SAS Studio to write and submit code in this workshop.
4. The Start Page tab is open by default, and it provides convenient links to start writing a program or use other point-and-click tools, including importing and querying data. Click **Program in SAS** to open a new tab with the Program Editor. Each item that you open in SAS Studio has its own tab.
5. The navigation pane is on the left side of SAS Studio. Each section in the navigation pane provides access to different resources in your SAS environment.
    - **Open Items** enables you to quickly view and access files that are currently open in your SAS Studio session. You can see that the list of open files matches the tabs.
    - **SAS Server** provides access to files and folders in your SAS environment. Keep in mind that the files and folders that you see here are not on your local machine. They're on the server where SAS is running. If you have the available permissions, you can use the **Upload files** button to load local files to the SAS server. The SAS Server also enables you to perform other tasks, such as background submit and schedule programs and basic file operations, such as copy, paste, and delete.
    - **Steps** enables you to build flows in SAS Studio. In a flow, you can build a sequence of steps, including programs, queries, and data transformations, as well as your own custom steps.
    - **Tasks** enables you to access tasks in SAS Studio. Tasks are based on SAS procedures and generate SAS code and formatted results for you.
    - **Snippets** enables you to access snippets. Snippets are lines of commonly used code or text that you can save and reuse. SAS Studio provides several predefined snippets, and you can create your own.
    - **Library Connections** provides access to your SAS libraries. You can open SAS tables and add them to your programs. You can even expand a table and view its columns.
    - **Git Repositories** enables you to access the basic Git features from within SAS Studio for maintaining and viewing a history of program edits.

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HOW Steps

Sample Results

6. The following results were generated using the free version of ChatGPT 5.2, and Claude Sonnet 4.6 in my logged in accounts. Note that LLM's are updated frequently. Use the following as a guide, not a set in stone solution. You may notice that depending on the generated response, prompts vary slightly.
7. ChatGPT 5.2 Results: https://chatgpt.com/share/69a74ffa-8bf8-8007-92c5-99c9d3472896
8. Claude Sonnet 4.6 Results: https://claude.ai/share/852a389e-9f8b-4171-99d8-540c4ef4774f

Demo 1: Building, Refining, and Documenting SAS Code

9. In **SAS Server**, expand **Files** > **Home** > **workshop** > **AI_Prog**. Double-click **Demo_01.sas**.
10. Demo 1 includes nine prompts to define the SAS environment, build, refine, and document a SAS program.
11. Click **New tab** +, **SAS program** to open a new SAS program. Paste the results from prompts in this program. Results will vary depending on the LLM you choose to use and the level of access you have.
12. Submit each prompt, one at a time. Copy and paste the code from the responses generated. Run each piece of code to validate accuracy. If an error is generated, copy and paste the error and ask the LLM what caused the error to prompt an accurate response.

Demo 2: Graphing and Navigating LLM Hallucinations

13. In **SAS Server**, expand **Files** > **Home** > **workshop** > **AI_Prog**. Double-click **Demo_02.sas**.
14. Demo 2 includes nine prompts to create a vertical bar chart displaying the number of cars for each value of Type. The prompts increase in complexity and may lead to hallucinations in the LLM.
15. Click **New tab** +, **SAS program** to open a new SAS program. Paste the results from prompts in this program. Results will vary depending on the LLM you choose to use and the level of access you have.
16. Submit each prompt, one at a time. Copy and paste the code from the responses generated. Run each piece of code to validate accuracy. If an error is generated, copy and paste the error and ask the LLM what caused the error to prompt an accurate response. If an invalid option is added, be sure to remove it from the code you are running.

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Demo 1 Code Solution

```
/*******************************************************/
/* SAS Programming with an AI Assistant                */
/* SAS Innovate 2026                                   */
/* Demo 1: Building, Refining, and Documenting SAS Code */
/*******************************************************/

/* Prompt 1:
I am coding in SAS using SAS Viya version 2025.12.
Please treat SAS as the default programming language for this
conversation.
When I ask for code:
 - Use Base SAS (DATA step, PROC SQL, PROCs)
 - Include brief explanations of what the code is doing
 - Minimize steps and use the most efficient methods when possible
 - Follow best practices for readability and teaching
*/

/* Prompt 2:
Provide a description of the CARS table in the SASHELP library.
What are the columns and basic attributes?
*/

proc contents data=sashelp.cars short varnum;
run;

/* Prompt 3:
Generate a PROC PRINT step that will print the first 20 rows of
SASHELP.CARS.
*/

proc print data=sashelp.cars(obs=20);
run;

/* Prompt 4:
Write a DATA step that computes the following new columns:
 - MPG_Avg: average of MPG_City and MPG_Highway
 - Doors: Search the Model column for the string '2dr'.
```

```
          If it is found, then assign the number 2 to Doors.
          Also search Model for the string 4dr.
          If it is found, then assign the number 4 to doors.
          Otherwise assign a missing value.
Keep the following columns: Make, Model, Type, Origin, MPG_Avg, Doors
*/

data cars_modified;
    set sashelp.cars;

    /* Compute average MPG (handles missing automatically) */
    MPG_Avg = mean(MPG_City, MPG_Highway);

    /* Determine number of doors from Model text */
    if find(Model, '2dr', 'i') then Doors = 2;
    else if find(Model, '4dr', 'i') then Doors = 4;
    else Doors = .;

    /* Keep only requested variables */
    keep Make Model Type Origin MPG_Avg Doors;
run;

/* Prompt 5:
Could you generate a similar result using PROC SQL?
*/

proc sql;
    create table cars_modified as
    select
        Make,
        Model,
        Type,
        Origin,

        /* Average MPG (mean ignores missing automatically in SQL) */
        mean(MPG_City, MPG_Highway) as MPG_Avg,

        /* Determine Doors from Model text */
        case
```

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```
            when find(Model, '2dr', 'i') > 0 then 2
            when find(Model, '4dr', 'i') > 0 then 4
            else .
        end as Doors

    from sashelp.cars;
quit;

/* Prompt 6:
Generate a report that counts the number of cars for each value of
Type.
*/

proc freq data=sashelp.cars;
    tables Type / nocum nopercent;
run;

/* Prompt 7:
How can I order the report by descending frequency? I would also like
to add a title.
*/

title "Number of Cars by Type (Ordered by Descending Frequency)";

/* Frequency report ordered by highest count first */
proc freq data=sashelp.cars order=freq;
    tables Type / nocum nopercent;
run;

/* Clear title (good practice) */
title;

/* Prompt 8:
Enhance the PROC FREQ step to create a macro variable that allows a
user to filter the report for a specific make of car.
*/

%let make_filter = Toyota;
```

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```
/* Add dynamic title */
title "Number of Cars by Type for &make_filter (Ordered by Descending
Frequency)";

/* Frequency report filtered by Make */
proc freq data=sashelp.cars(where=(Make="&make_filter")) order=freq;
    tables Type / nocum nopercent;
run;

/* Clear title */
title;

/* Prompt 9:
Here is my final program.
Insert brief comments inline to explain each statement.
Additionally, write a simple summary of the entire program that I can
use for documentation.
*Copy and paste full program*
*/
/* Answers will vary */
```

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Demo 2 Code Solution

```
/*****************************************************/
/* SAS Programming with an AI Assistant              */
/* SAS Innovate 2026                                 */
/* Demo 2: Graphing and Navigating LLM Hallucinations */
/*****************************************************/

/* Prompt 1:
Create a vertical bar chart that displays the number of cars for each
value of Type.
*/

/* Add title */
title "Number of Cars by Type";

/* Vertical bar chart of car counts by Type */
proc sgplot data=sashelp.cars;
    vbar Type / stat=freq;   /* Compute and display frequency counts
*/
run;

/* Clear title */
title;

/* Prompt 2:
Can you order the bars by descending frequency?
*Desired response: categoryorder=respdesc on the VBAR statement
*/

title "Number of Cars by Type (Ordered by Descending Frequency)";

proc sgplot data=sashelp.cars;
    vbar Type / stat=freq categoryorder=respdesc;
run;

title;

/* Prompt 3:
```

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```
Can you add data labels to each bar?
*/

title "Number of Cars by Type";

/* Vertical bar chart with frequency labels */
proc sgplot data=sashelp.cars;
    vbar Type /
        stat=freq
        datalabel categoryorder=respdesc;   /* Display count on each
bar */
run;

/* Clear title */
title;

/* Prompt 4:
How can I make each bar a different color?
*Desired response: group=Type on the VBAR statement and
                   STYLEATTRS datacolors=(hex-color-list) statement
*/

title "Number of Cars by Type (Ordered by Descending Frequency)";

proc sgplot data=sashelp.cars;
    styleattrs datacolors=(cx4E79A7 cxF28E2B cxE15759 cx76B7B2
cx59A14F cxEDC948);

    vbar Type /
        stat=freq
        categoryorder=respdesc
        group=Type
        groupdisplay=cluster
        datalabel;
run;

title;

/* Prompt 5:
```

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```
Can you make a color scheme for shades of dark to light pink?
*/

/* Add title */
title "Number of Cars by Type (Ordered by Descending Frequency)";

/* Pink gradient color scheme */
proc sgplot data=sashelp.cars;

    /* Dark → Light Pink */
    styleattrs datacolors=(
        cx8B004B   /* Dark magenta-pink */
        cxC2185B   /* Deep pink */
        cxE91E63   /* Medium pink */
        cxF06292   /* Light pink */
        cxF8BBD0   /* Very light pink */
        cxFCE4EC   /* Pale blush */
    );

    vbar Type /
        stat=freq
        categoryorder=respdesc
        group=Type
        groupdisplay=cluster
        datalabel;
run;

/* Prompt 6:
How can I remove the outline around each bar?
Desired response: nooutline on the VBAR statement
*/

/* Add title */
title "Number of Cars by Type (Ordered by Descending Frequency)";

/* Pink gradient color scheme */
proc sgplot data=sashelp.cars;

    /* Dark → Light Pink */
```

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```
    styleattrs datacolors=(
        cx8B004B   /* Dark magenta-pink */
        cxC2185B   /* Deep pink */
        cxE91E63   /* Medium pink */
        cxF06292   /* Light pink */
        cxF8BBD0   /* Very light pink */
        cxFCE4EC   /* Pale blush */
    );

    vbar Type /
        stat=freq
        categoryorder=respdesc
        group=Type
        groupdisplay=cluster
        datalabel
        nooutline;
run;

title;

/* Prompt 7:
Add a macro variable to the step to filter the bar chart for a
specified value of Origin.
*/

/* User-specified Origin */
%let origin_filter = Asia;

title "Number of Cars by Type for &origin_filter (Ordered by
Descending Frequency)";

proc sgplot data=sashelp.cars(
               where=(Origin="&origin_filter")
            );

    /* Dark  Light Pink gradient */
    styleattrs datacolors=(
        cx8B004B
        cxC2185B
```

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```
        cxE91E63
        cxF06292
        cxF8BBD0
        cxFCE4EC
    );

    vbar Type /
        stat=freq
        categoryorder=respdesc
        group=Type
        groupdisplay=cluster
        datalabel
        nooutline;
run;

title;   /* Clear title */

/* Prompt 8:
Make the filter case insensitive.
*/

/* User input (can be any case) */
%let origin_filter = usa;

/* Proper-case version for display in title */
%let origin_display = %sysfunc(propcase(&origin_filter));

title "Number of Cars by Type for &origin_display (Ordered by
Descending Frequency)";

proc sgplot data=sashelp.cars(
              where=(upcase(Origin)=upcase("&origin_filter"))  /*
Case-insensitive filter */
            );

    /* Dark  Light Pink gradient */
    styleattrs datacolors=(
        cx8B004B
        cxC2185B
```

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```
        cxE91E63
        cxF06292
        cxF8BBD0
        cxFCE4EC
    );

    vbar Type /
        stat=freq
        categoryorder=respdesc
        group=Type
        groupdisplay=cluster
        datalabel
        nooutline;   /* Remove borders */

run;

title;   /* Clear title */

/* Prompt 9:
Define a macro and use macro conditional processing.
If the value of the macro variable is usa, then UPCASE the value in
the title.
Else, PROPCASE the value.
If the value is not Asia, USA, or Europe, write a note to the log that
the valid values are Asia, Europe, or USA in any case.
If no value is provided, create the bar chart for all origins and
write a note to the log.
*/

/* Define macro */
%macro origin_chart(origin_filter);

    /* Check if parameter is missing */
    %if %length(&origin_filter) = 0 %then %do;

        %put NOTE: No origin_filter specified. Chart generated for ALL
origins.;
```

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```
        title "Number of Cars by Type for All Origins (Ordered by
Descending Frequency)";

        proc sgplot data=sashelp.cars;

            styleattrs datacolors=(
                cx8B004B
                cxC2185B
                cxE91E63
                cxF06292
                cxF8BBD0
                cxFCE4EC
            );

            vbar Type /
                stat=freq
                categoryorder=respdesc
                group=Type
                groupdisplay=cluster
                datalabel
                nooutline;

        run;

        title;

    %end;

    %else %do;

        /* Standardize for validation */
        %let origin_check = %upcase(&origin_filter);

        /* Validate Origin value */
        %if &origin_check = USA or
            &origin_check = ASIA or
            &origin_check = EUROPE %then %do;

            /* Title formatting logic */
```

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```
            %if &origin_check = USA %then %do;
                %let origin_display = %upcase(&origin_filter);
            %end;
            %else %do;
                %let origin_display =
%sysfunc(propcase(&origin_filter));
            %end;

            title "Number of Cars by Type for &origin_display (Ordered
by Descending Frequency)";

            proc sgplot data=sashelp.cars(
                           where=(upcase(Origin)="&origin_check")
                        );

                styleattrs datacolors=(
                    cx8B004B
                    cxC2185B
                    cxE91E63
                    cxF06292
                    cxF8BBD0
                    cxFCE4EC
                );

                vbar Type /
                    stat=freq
                    categoryorder=respdesc
                    group=Type
                    groupdisplay=cluster
                    datalabel
                    nooutline;
            run;

            title;

        %end;

        %else %do;
```

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```
            %put NOTE: Valid values for origin_filter are Asia,
Europe, or USA (any case).;
            %put NOTE: Chart was not generated.;
        %end;

    %end;

%mend origin_chart;

%origin_chart();
%origin_chart(asia);
%origin_chart(usa);
%origin_chart(Europe);
%origin_chart(africa);
```

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HOW Slides

As you all have seen, AI is reshaping how work gets done across every industry. Whether we choose to engage with it or not, it's influencing the tools, platforms, and expectations around us. Understanding how to leverage AI is no longer optional- it's a professional responsibility. There are several large language models (or LLMs)- the key players are Copilot, Gemini, Claude, and ChatGPT.Today, we're going to explore how to use ChatGPT as a partner in developing code to inspire ways that it may

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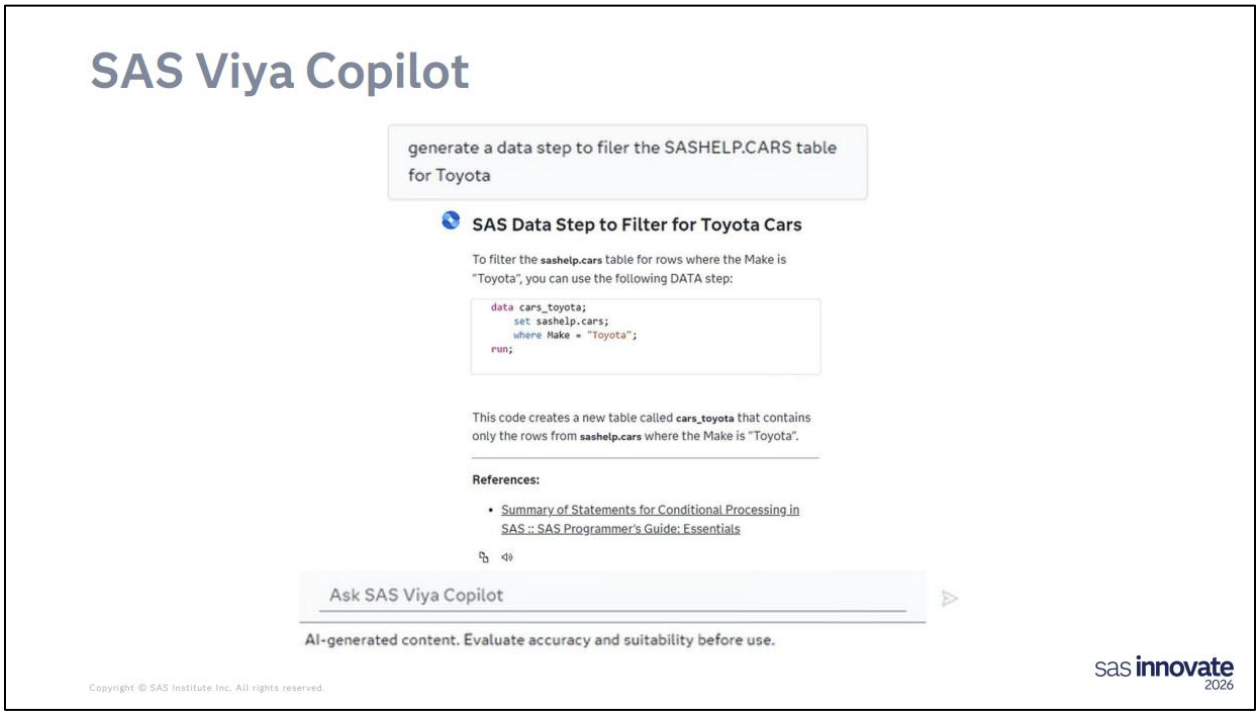
enhance workflows. But first, let's discuss SAS Studio Copilot and break down where the key LLM's shine.



SAS Viya Copilot transforms prompts into analytical actions, recommendations, and explanations. In SAS Studio, you can use SAS Viya Copilot to answer questions, generate code, explain or resolve errors and more. In this example, I asked Copilot to generate a DATA step to filter the SASHELP.CARS dataset for Toyota. It generated a DATA step, referenced documentation, and allowed me to add it directly in my program where my cursor was.

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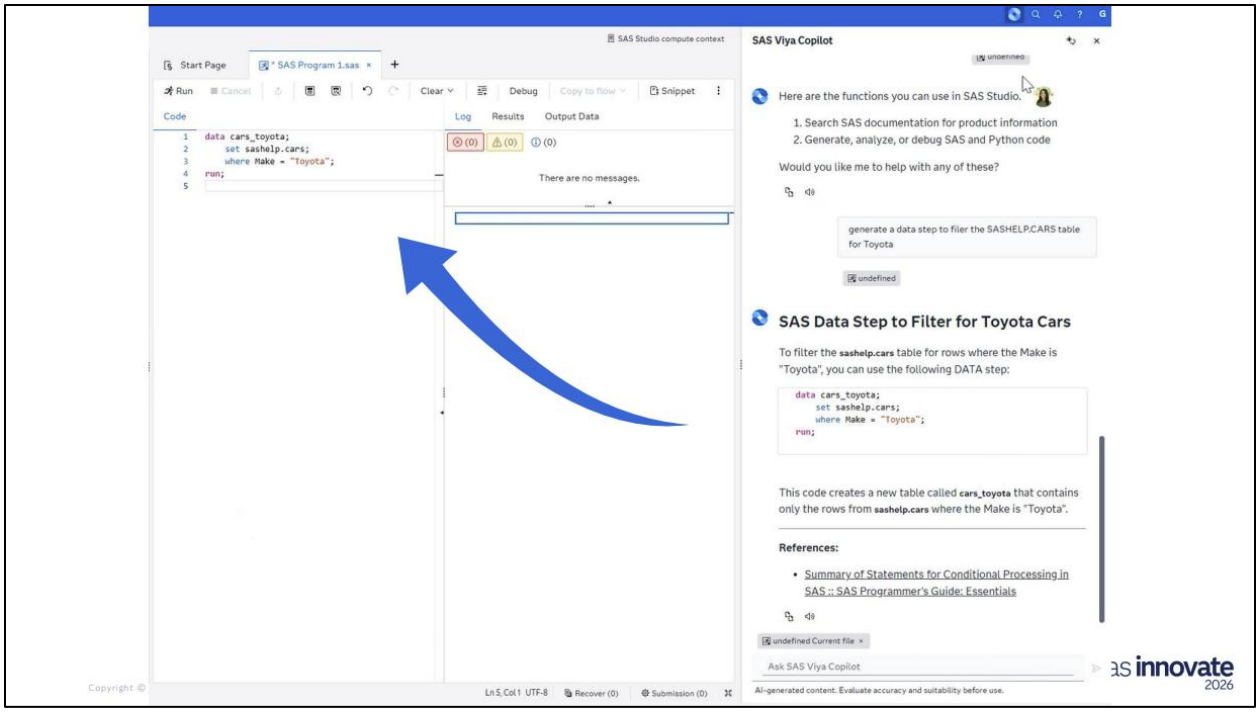
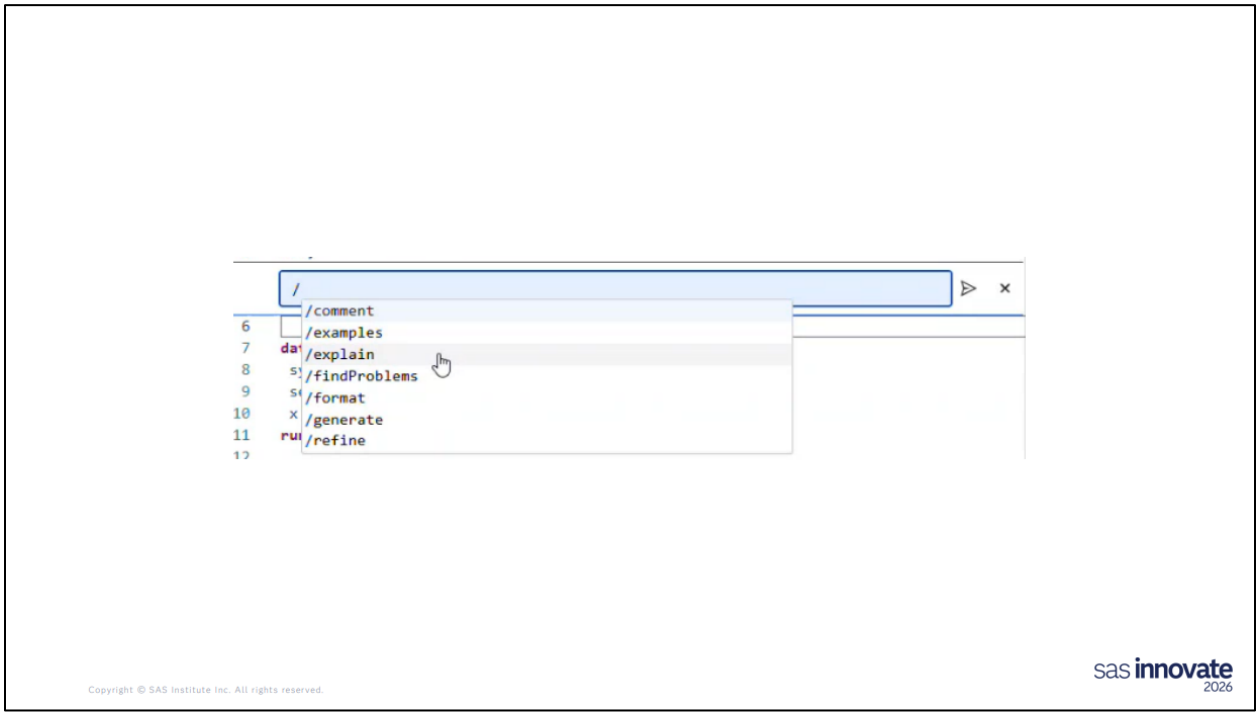


Here's what it looks like in the SAS Studio application.



Additionally, you can use Copilot inline to add comments, generate examples, explain code, find issues, and format, generate, or refine code.

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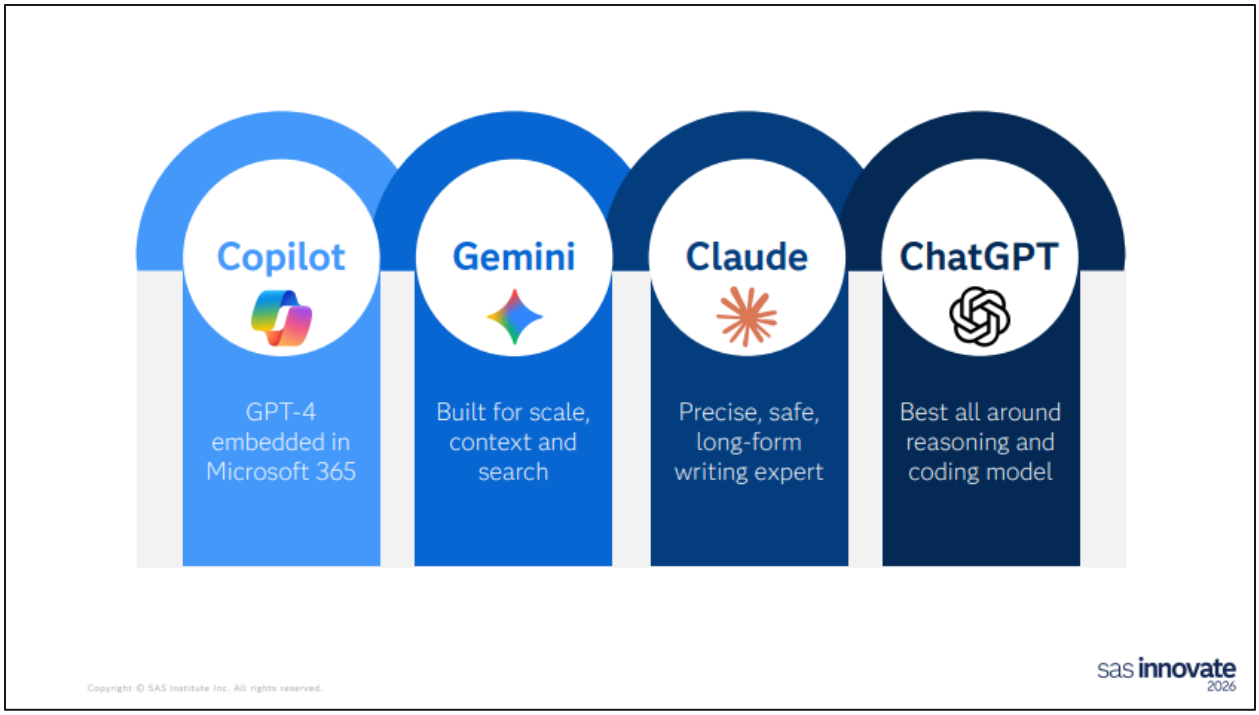

Microsoft Copilot GPT-4 embedded in Microsoft 365. Drafts in Word, analyzes in Excel, builds decks in PowerPoint, and assists developers via GitHub Copilot.

Google Gemini Built for scale and context. Leads in multimodality, large document analysis, and Google Workspace productivity. Best for information retrieval and enterprise tasks.

Anthropic Claude Optimized for precision and safety. Leads in long-form writing, legal/policy/academic documents, and low-hallucination environments.

OpenAI ChatGPT Best all-around model. Leads in reasoning, coding, tool calling, and agentic workflows. Top choice for complex technical and business tasks.

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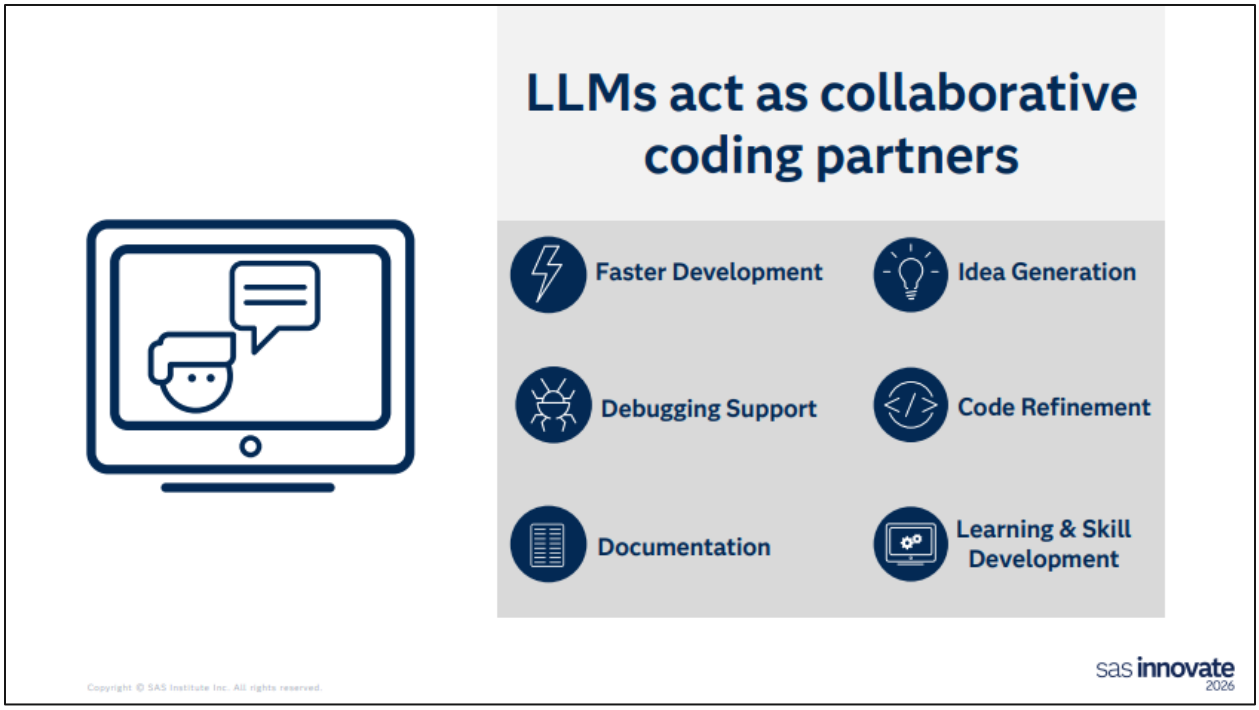

An LLM acts as a **collaborative coding partner**—accelerating routine work, supporting deeper thinking, and helping you focus more on design and problem-solving rather than syntax and recall.

1. Faster Development: LLMs can generate code, standard procedures, and common patterns quickly-less repetitive typing and faster starts.

2. Idea Generation & Problem Solving: They help brainstorm approaches, alternative methods, and surface built-in functions or techniques you may not immediately recall.

3. Debugging Support: Analyze error messages, explain likely causes, and propose fixes helping you identify problems faster.

4. Code Refinement: Restructure code for readability, efficiency, or best practices, improving maintainability and performance.

5. Documentation & Explanation: Generate comments, write technical documentation, and explain complex logic in plain language, which is helpful for knowledge sharing.

6. Learning & Skill Building: Serve as an interactive tutor—explaining concepts, comparing approaches, and reinforcing best practices as you code.

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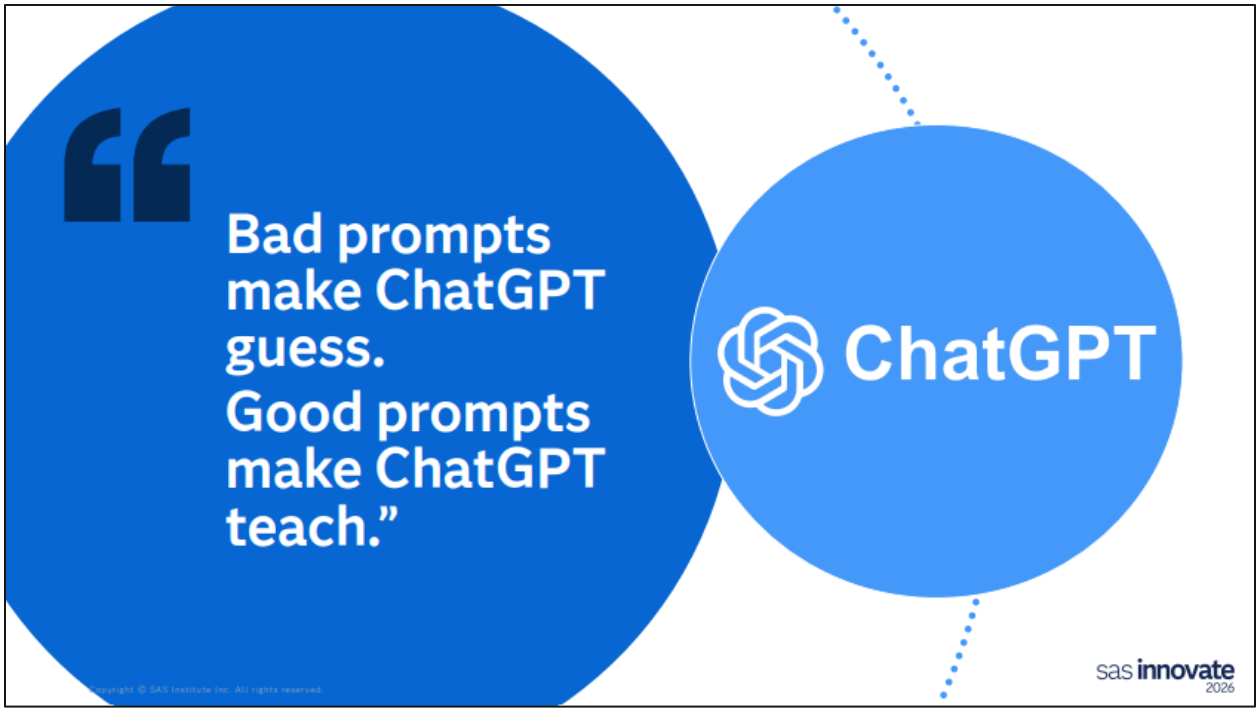
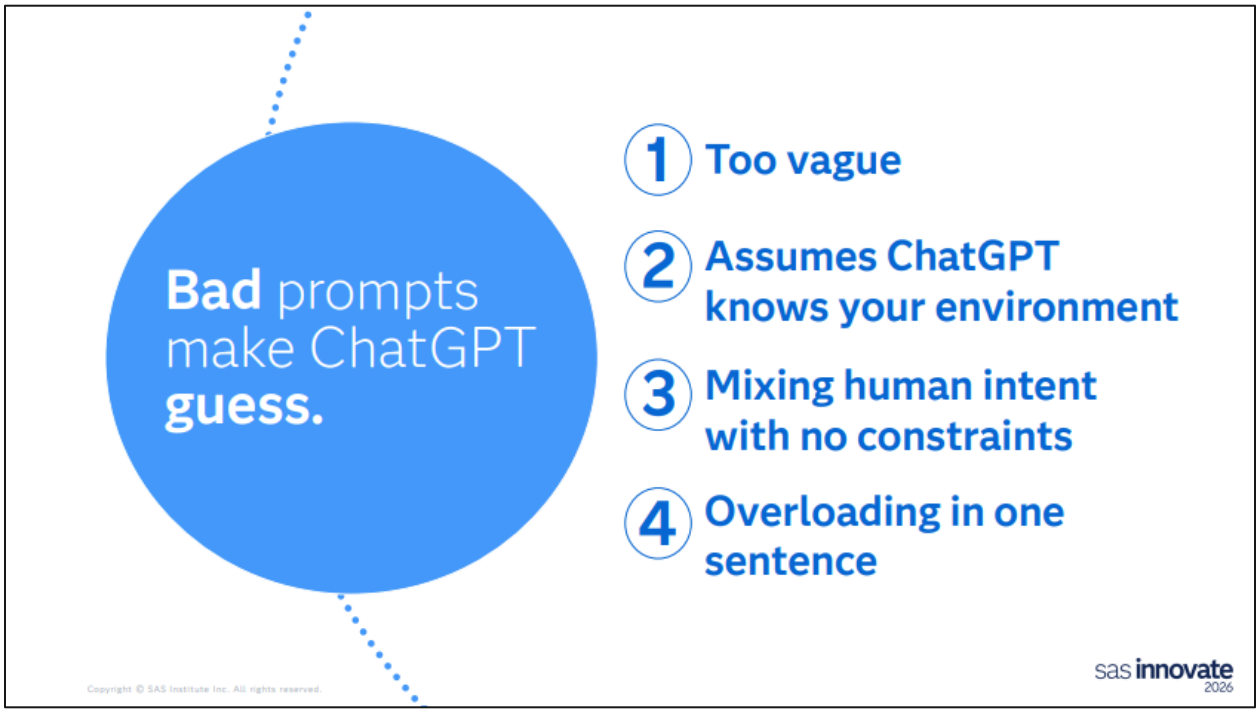


When working with LLM's, the way you write a prompt determines the answer an LLM generates and allows users to customize output. Writing sound prompts is known as "prompt engineering" and can vary by AI tool. This is a quote ChatGPT gave me: Bad prompts make ChatGPT guess. Good prompts make ChatGPT teach. And this is true. As we use ChatGPT to code, we should view it as an "assistant" and approach solutions provided with a responsible amount of caution.



1. How do I do this in code?

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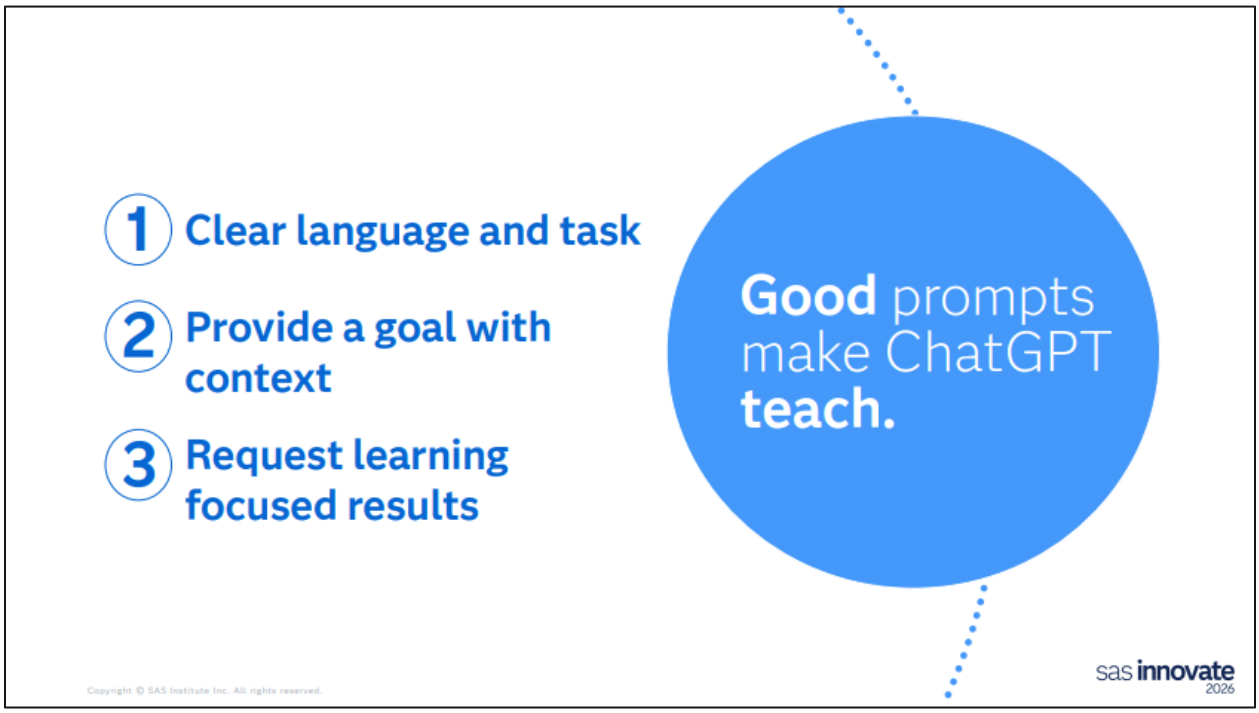
2. Read a CSV and summarize it.
3. I want to clean some data and make it better.
4. I have data with names and prices and reviews and I want to analyze it and visualize it and make it ready for a model.



1. I want to count the number of unique values in a column. What's the best way to do this?
2. I have a dataset with product names and prices. How can I calculate average price per product?
3. Show a simple example of summarizing data and explain the logic step by step.

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PROC SGPLOT Hallucinations

Ahhh — yes. PROC SGPLOT is where weak models completely fall apart.

They:

- Hallucinate options
- Confuse `STYLEATTRS` vs `FILLATTRS`
- Misuse `GROUP=`
- Forget required statements
- Mix GTL syntax into SGPLOT
- Invent options that don't exist

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Before we begin, it's important to know that LLMs can hallucinate, meaning it gives you something that is incorrect, or even something that doesn't exist. We will likely see this in the demo and you'll notice that the LLM responds with a hallucinated answer in the same confident tone as a correct answer. After I created the demo, and tested with the free version of ChatGPT, I frequently got varying results when working with PROC SGPLOT. I asked ChatGPT why this was and it recognized its own flaws! Keep these in mind as we go through the demo and know that if you log in or use a paid version of ChatGPT, you may not run into these issues.

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